



tronics equipment designers. These have always helped in delivering the best products; the ease of testing equipment with computer systems is only expected to improve productivity. This also minimises the chances of human errors in the final result, as the signal generated can be calibrated and values can be automated via the computer.

However, accuracy is not enough. "Customers expect higher precision than in the past, that is, 16-bit amplitude resolution instead of 14-bit used in the actual generation," says Rao.

V. Mahadevan, managing director, VigVen Tech Mark Pvt Ltd, supports this argument about high-resolution equipment. He says, "Such precision would result in minute changes being highlighted, resulting in better end devices."

Rao mentions another requirement as, "requests for digital pattern generators as additional functionality." With the move to digital signals, this was bound to happen sooner or later. Agilent, NI and Tektronix, among others, provide products catering to this category as well.

Signal generation from a single clock

Newer function generators, known as arbitrary waveform generators (AWGs), employ direct digital synthesisers (DDSes) in signal generation. DDSes can be used for creating arbitrary waveforms from a single, fixed-frequency reference clock.

Use of DDS has been another advantage in coming up with high-frequency devices. A DDS has many advantages over the earlier phase-locked loop (PLL), including much better frequency agility, improved phase noise and precise control of the output phase across frequency-switching transitions.

Trueform technology from Keysight offers standard features along with baseband I-Q player, 80MHz and 120MHz with one and two channels, two-channel coupling and

Hack for signal generation

An interesting approach for signal generation is the use of a sound card fitted on your personal computer system to output the required wave. Within certain specified frequency ranges and amplitudes, you can generate your very own personal frequencies. Employing some software instructions, it should be a good DIY.

Test it on Cloud

To make testing easy for engineers, a trend towards incorporating multiple equipment in a single equipment is being seen. Kicking it up a notch, you have Cloud testing platforms from Advantest. The capability to test semiconductor chips online creates an ease in measurements. Now the tester can plug in the probes and let the Cloud take care of the rest. All updates in the instruments are done at CTS end and the user can work comfortably simply by paying a monthly fee. Additionally, instead of using several different components and getting confused between readings, a single platform takes care of multiple tests.

synchronisation, Trueform waveform sequencing and PRBS series patterns. "Trueform technology offers more capability, fidelity and flexibility," says Kishore S., application engineer, Keysight Technologies India Pvt Ltd. 33600A and 33500B series Trueform waveform generators from Keysight feature this technology.

Flexible equipment design

Rao says, "Our preferred implementation is based on FPGAs containing several DDSes, thus providing maximum flexibility at reasonable prices. It allows us to tailor the instrument to the needs of big-volume customers."

Today, most function generators adopt a digital approach. Analogue function generators are still used as these can provide levels of performance that are more than satisfactory for exact requirements needed in many development laboratories. Also, analogue function generators tend to be much cheaper.

Oh, the challenges

"Some test challenges are generating a waveform with many points, simulating signals with high integrity, synchronising two signals to generate pseudorandom binary sequence signals, creating differential signals, changing one segment of arb without redesigning the whole signal and needing to sweep arb through a set of different frequencies," says Kishore.



Cloud testing

Another matter of concern with function generators is the price. "Price is, as usual, an issue," adds Rao. However, with the release of multiple test equipment incorporated in a single device, this should not be a major issue now. Bringing the best in technology also has its problems that can correlate to cost. "The time to market causes the product to be designed with the latest processors to be delayed," says Mahadevan.

Could there be a wave of change

Addition of multiple testers in a single piece of equipment has been the manufacturing strategy for some time now. When you incorporate multiple equipment with a familiar computer interface, working with the equipment will be highly simplified.

Looking forward, we can surely expect some other integration into the function generator. If the cost challenges can be somehow handled, a function generator can be regularly updated. And it would not just lie on the corner table for an entire year to be used only once in a while. **EFY**